

TAREA NoSQL Máster UCM Data Science, Big Data & Business Analytics 2024-2025

Ejercicio nº1

```
db.movies.find()
```

	_id *	title	year	cast	genres
9	67a0da856a8798b4562ed624	Searching Ruins on Broadway, Galveston, f	1900	Array[1]	Array[1]
10	67a0da856a8798b4562ed625	The Tribulations of an Amateur Photograph	1900	Array[1]	Array[1]
11	67a0da856a8798b4562ed626	Trouble in Hogan's Alley	1900	Array[1]	Array[1]
12	67a0da856a8798b4562ed627	Boarding School Girls' Pajama Parade	1900	Array[1]	Array[1]
13	67a0da856a8798b4562ed628	Two Old Sparks	1900	Array[1]	Array[1]
14	67a0da856a8798b4562ed629	The Wonder, Ching Ling Foo	1900	Array[1]	Array[1]
15	67a0da856a8798b4562ed62a	Watermelon Contest	1900	Array[1]	Array[1]
16	67a0da856a8798b4562ed62b	An Affair of Honor	1901	Array[1]	Array[1]
17	67a0da856a8798b4562ed62c	Another Job for the Undertaker	1901	Array[1]	Array[1]

Ejercicio nº2

```
db.movies.countDocuments()
```

0.014 s	JSON
1	28795

Ejercicio nº3

```
db.movies.insertOne({
  title: "The Departed",
  year: 2006,
  cast: ["Leonardo DiCaprio", "Matt Damon", "Mark Wahlberg"],
  genres: ["police", "thriller"]
})
```

0.024 s

Tree

Key	Value	Type
(1)	{ acknowledged : true, insertedId : ObjectId("67a339006e68b4b1f67ac564") }	Object
acknowledged	true	Bool
insertedId	67a339006e68b4b1f67ac564	ObjectId

Ejercicio nº4

```
db.movies.deleteOne({ title: "The Departed" })
```

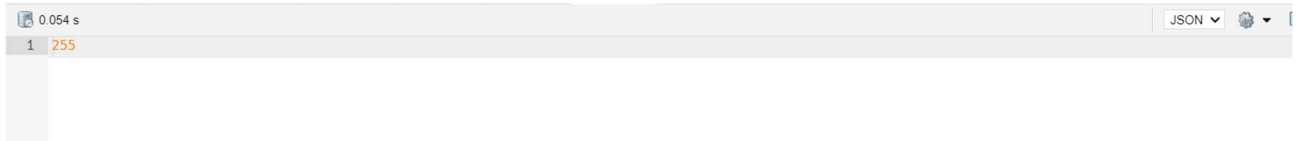
0.051 s

Tree

Key	Value	Type
(1)	{ acknowledged : true, deletedCount : 1 }	Object
acknowledged	true	Bool
deletedCount	1	Int32

Ejercicio nº5

```
db.movies.countDocuments({ cast: { $regex: "\\band\\b", $options: "i" } })
```

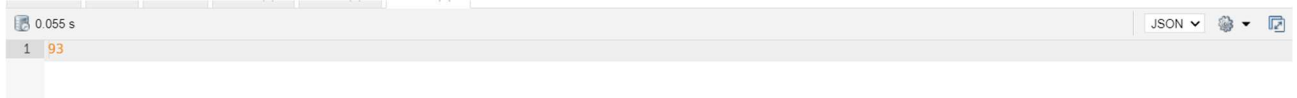


0.054 s

1	255
---	-----

JSON

```
db.movies.countDocuments({ cast: { $regex: "^and$", $options: "i" } })
```



0.055 s

1	93
---	----

JSON

- **La primera consulta** encuentra **todos los documentos** en los que "and" es una palabra separada **dentro** del valor de cast.
- **La segunda consulta** encuentra **solo** los documentos en los que cast es **exactamente** "and", sin ningún otro carácter.

Por lo tanto, la primera consulta dará un **número mayor de resultados** que la segunda, porque es más permisiva y encuentra "and" como una palabra completa dentro de frases más largas.

La mas correcta es la segunda consulta

Ejercicio nº6

```
db.movies.updateMany(  
  { cast: { $regex: "^and$", $options: "i" } },  
  { $pull: { cast: { $regex: "^and$", $options: "i" } } }  
)
```

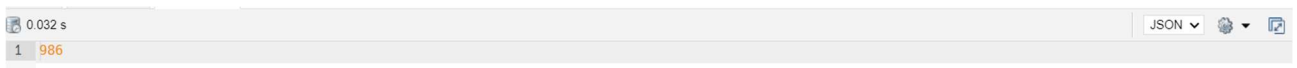


0.053 s

Key	Value	Type
(1)	{ acknowledged : true, matchedCount : 93, modifiedCount : 93 }	Object
acknowledged	true	Bool
matchedCount	93	Int32
modifiedCount	93	Int32

Ejercicio nº7

```
db.movies.countDocuments({ cast: { $size: 0 } })
```



0.032 s

1	986
---	-----

JSON

Ejercicio nº8

0.047 s			Tree		
Key	Value	Type			
1 (1)	{ acknowledged : true, matchedCount : 986, modifiedCount : 986 }	Object			
acknowledged	true	Bool			
matchedCount	986	Int32			
modifiedCount	986	Int32			

db.movies.find()

Edit Element - _id: ObjectId("67a49d1d63763eda51519438")

Parent
(document)

Type
Array

Add this field to
Current document only

Value
1 ["Undefined"]

Preview Code... Execute (Ctrl+*) Cancel

Ejercicio nº9

db.movies.countDocuments({ genres: { \$size: 0 } })

0.015 s		JSON		
1	901			

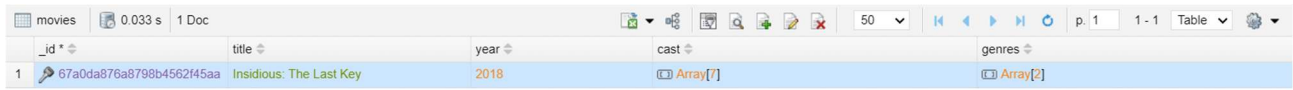
Ejercicio nº10

```
db.movies.updateMany(  
  { genres: { $size: 0 } },  
  { $set: { genres: ["Undefined"]} }  
)
```

0.044 s			Tree		
Key	Value	Type			
1 (1)	{ acknowledged : true, matchedCount : 901, modifiedCount : 901 }	Object			
acknowledged	true	Bool			
matchedCount	901	Int32			
modifiedCount	901	Int32			

Ejercicio nº 11

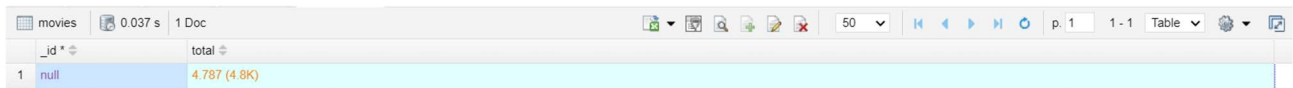
```
db.movies.find().sort({ year: -1 }).limit(1).pretty()
```



	_id	title	year	cast	genres
1	67a0da876a8798b4562f45aa	Insidious: The Last Key	2018	Array[7]	Array[2]

Ejercicio nº 12

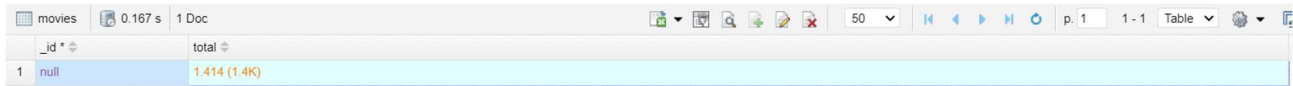
```
db.movies.aggregate([  
  { $match: { year: { $gte: 1999, $lte: 2018 } } },  
  { $group: { _id: null, total: { $sum: 1 } } }  
]);
```



	_id	total
1	null	4.787 (4.8K)

Ejercicio nº 13

```
db.movies.aggregate([  
  { $match: { year: { $gte: 1960, $lte: 1969 } } },  
  { $group: { _id: null, total: { $sum: 1 } } }  
]);
```



	_id	total
1	null	1.414 (1.4K)

Ejercicio nº 14

```
db.movies.aggregate([  
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } }  
]);  
  
db.movies.aggregate([  
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } },  
  { $sort: { total_pelis: 1 } }  
]);  
  
db.movies.aggregate([  
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } },  
  { $sort: { total_pelis: 1 } },
```

```
{ $limit: 1 }
```

```
]);
```

	_id *	total_pelis
1	1919	634

Ejercicio nº 15

```
db.movies.aggregate([
```

```
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } }
```

```
]);
```

```
db.movies.aggregate([
```

```
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } },
```

```
  { $sort: { total_pelis: 1 } }
```

```
]);
```

```
db.movies.aggregate([
```

```
  { $group: { _id: "$year", total_pelis: { $sum: 1 } } },
```

```
  { $sort: { total_pelis: 1 } },
```

```
  { $limit: 1 }
```

```
]);
```

	_id *	total_pelis
1	1906	7

Ejercicio nº 16

```
db.movies.aggregate([
```

```
  { $unwind: "$cast" },
```

```
  { $project: { _id: 0, actor: "$cast", movie: "$title", year: "$year" } },
```

```
  { $out: "actors" }
```

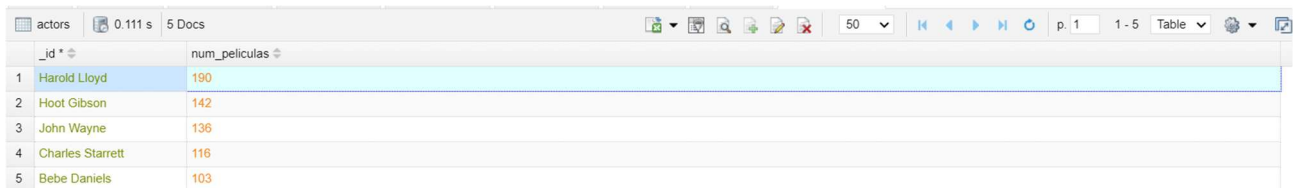
```
])
```

```
db.actors.countDocuments()
```

0.807 s		JSON			
1	83215				

Ejercicio nº 17

```
db.actors.aggregate([
  { $match: { actor: { $ne: "Undefined" } } },
  { $group: { _id: "$actor", num_peliculas: { $sum: 1 } } },
  { $sort: { num_peliculas: -1 } },
  { $limit: 5 }
])
```

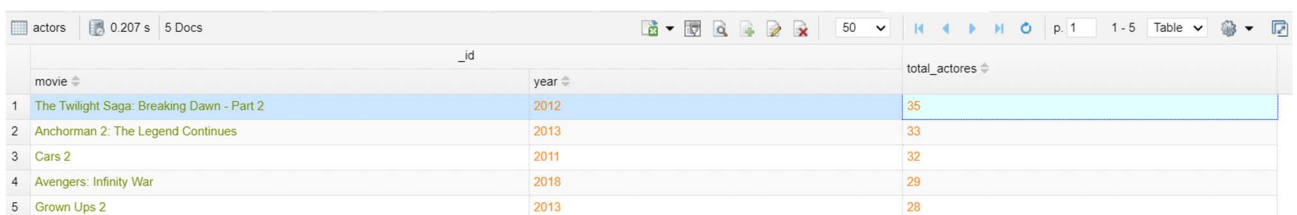


A screenshot of the MongoDB query results interface. The top bar shows 'actors' collection, '0.111 s' execution time, and '5 Docs' returned. The table has two columns: '_id' and 'num_peliculas'. The results are sorted by 'num_peliculas' in descending order.

	_id	num_peliculas
1	Harold Lloyd	190
2	Hoot Gibson	142
3	John Wayne	136
4	Charles Stanrett	116
5	Bebe Daniels	103

Ejercicio nº 18

```
db.actors.aggregate([
  { $group: {
    _id: { movie: "$movie", year: "$year" },
    total_actores: { $sum: 1 }
  } },
  { $sort: { total_actores: -1 } },
  { $limit: 5 }
])
```



A screenshot of the MongoDB query results interface. The top bar shows 'actors' collection, '0.207 s' execution time, and '5 Docs' returned. The table has three columns: 'movie', 'year', and 'total_actores'. The results are sorted by 'total_actores' in descending order.

	movie	year	total_actores
1	The Twilight Saga: Breaking Dawn - Part 2	2012	35
2	Anchorman 2: The Legend Continues	2013	33
3	Cars 2	2011	32
4	Avengers: Infinity War	2018	29
5	Grown Ups 2	2013	28

Ejercicio nº 19

```
db.actors.aggregate([
  { $match: { actor: { $ne: "Undefined" } } },
  { $group: { _id: "$actor", inicio: { $min: "$year" }, fin: { $max: "$year" } } },
  { $project: { actor: "$_id", inicio: 1, fin: 1, años_trabajados: { $subtract: ["$fin", "$inicio"] } } },
  { $sort: { años_trabajados: -1 } },
  { $limit: 5 }
]);
```

actors 0.136 s 5 Docs				
	_id	inicio	fin	actor
1	Harrison Ford	1.919 (1.9K)	2.017 (2.0K)	Harrison Ford
2	Gloria Stuart	1.932 (1.9K)	2.012 (2.0K)	Gloria Stuart
3	Lillian Gish	1.912 (1.9K)	1.987 (2.0K)	Lillian Gish
4	Kenny Baker	1.937 (1.9K)	2.012 (2.0K)	Kenny Baker
5	Mickey Rooney	1.932 (1.9K)	2.006 (2.0K)	Mickey Rooney

Ejercicio nº 20

```
db.actors.aggregate([
  { $unwind: "$genres" },
  { $project: { _id: 0, genres: "$genres", actor: "$actor", movie: "$movie", year: "$year" } },
  { $out: "genres" }
])
```

```
db.genres.countDocuments()
```

Ejercicio nº 21

```
db.genres.aggregate([
  { $group: { _id: { year: "$year", genre: "$genres" }, totalMovies: { $addToSet: "$movie" } } },
  { $project: { year: "$_id.year", genre: "$_id.genre", totalMoviesCount: { $size: "$totalMovies" } } },
  { $sort: { totalMoviesCount: -1 } },
  { $limit: 5 }
])
```

genres 0.239 s 5 Docs				
	year	genre	year	totalMoviesCount
1	1919	Drama	1919	291
2	1925	Drama	1925	247
3	1924	Drama	1924	233
4	1919	Comedy	1919	226

Ejercicio nº 22

```
db.genres.aggregate([
  { $match: { actor: { $ne: "Undefined" } } },
  { $group: { _id: "$actor", uniqueGenres: { $addToSet: "$genres" } } },
```

```
{ $project: { actor: "$_id", genres: "$uniqueGenres", genreCount: { $size: "$uniqueGenres" } } },
{ $sort: { genreCount: -1 } },
{ $limit: 5 }
])
```

genres 0.399 s 5 Docs			
	_id *	actor	genres
1	Dennis Quaid	Dennis Quaid	Array[20]
2	James Mason	James Mason	Array[19]
3	Matt Damon	Matt Damon	Array[19]
4	Michael Caine	Michael Caine	Array[19]
5	Gene Hackman	Gene Hackman	Array[18]

Ejercicio nº 23

```
db.genres.aggregate([
{ $group: { _id: { movie: "$movie", year: "$year" }, uniqueGenres: { $addToSet: "$genres" } } },
{ $project: { movie: "$_id.movie", year: "$_id.year", genres: "$uniqueGenres", genreCount: { $size:
"$uniqueGenres" } } },
{ $sort: { genreCount: -1 } },
{ $limit: 5 }
])
```

_id					
	movie	year	movie	year	genres
1	American Made	2017	American Made	2017	Array[7]
2	Dunkirk	2017	Dunkirk	2017	Array[6]
3	Wonder Woman	2017	Wonder Woman	2017	Array[6]
4	The Dark Tower	2017	The Dark Tower	2017	Array[6]
5	My Little Pony: The Movie	2017	My Little Pony: The Movie	2017	Array[6]

Ejercicio libre nº1

Aquí tenemos una consulta ejecutada en la colección genres para obtener los actores que han actuado en más películas del mismo género, mostrando el género, el actor y el número de películas en ese género.

```
db.genres.aggregate([
{ $match: { actor: { $ne: "Undefined" } } },
{ $group: { _id: { actor: "$actor", genre: "$genres" }, movieCount: { $sum: 1 } } },
{ $sort: { movieCount: -1 } },
{ $limit: 5 }
])
```


	actor	_id	genre	movieCount
1	Harold Lloyd		Comedy	186
2	Hoot Gibson		Western	131
3	Charles Starrett		Western	103
4	Johnny Mack Brown		Western	86
5	Tom and Jerry		Animated	85

Ejercicio libre nº 2

Aquí tenemos una consulta basada en el pipeline de agregación que encuentra a los actores con el mayor número de películas en absoluto, independientemente del género.

```
db.genres.aggregate([
  { $match: { actor: { $ne: "Undefined" } } },
  { $group: { _id: "$actor", movieCount: { $addToSet: "$movie" } } },
  { $project: { actor: "$_id", totalMovies: { $size: "$movieCount" } } },
  { $sort: { totalMovies: -1 } },
  { $limit: 5 }
])
```

genres		0.490 s	5 Docs	    		50	   		p. 1	1 - 5	Table			
	_id	actor	totalMovies											
1	Harold Lloyd	Harold Lloyd	188											
2	Hoot Gibson	Hoot Gibson	142											
3	John Wayne	John Wayne	136											
4	Charles Starrett	Charles Starrett	115											
5	Bebe Daniels	Bebe Daniels	103											